

Closed forms and sharp minima of ωt_c on three classical point-vortex collapse families

Chaos

GENChase. github.com/SharpMeow/GENChase

First public: Hendrick's Identity 19 September 2026; parallelogram lock and quincunx lock 20 September 2026

A note. The motion on each family is classical. The closed product and the unique algebraic floor are not.

Abstract.

Self-similar collapse of N point vortices is classical. Gröbli (1877) and Novikov–Sedov (1979) found the families; Aref (2010) and Gotoda (2020) wrote the collapse rate A and the spin B as separate functions of a shape angle. Their dimensionless product $\omega_0 t_c = -\frac{B}{2A}$ is the pitch of the logarithmic spiral the configuration traces as it shrinks. This note records three closed forms of that product, each with a unique algebraic interior minimum that the source papers do not state:

- Hendrick's Identity, on Gröbli's $\Gamma = (1, 1, -\frac{1}{2})$ family: $\omega_0 t_c = \frac{2-\cos^2\theta}{\sin(2\theta)} \geq \sqrt{2}$, equality at $\tan\theta = \frac{1}{\sqrt{2}}$.
- The parallelogram lock, on the Novikov–Sedov four-vortex parallelogram: $\omega_0 t_c = \left(\frac{\sqrt{3}}{4}\right) \frac{4-\cos 2\theta}{\sin(2\theta)} \geq 3\frac{\sqrt{5}}{4}$, equality at $\cos 2\theta = \frac{1}{4}$.
- The quincunx lock, on the Novikov–Sedov five-vortex quincunx: $\omega_0 t_c = \left(\frac{3}{16}\right) \frac{7-4\cos 2\theta}{\sin(2\theta)} \geq 3\frac{\sqrt{33}}{16}$, equality at $\cos 2\theta = \frac{4}{7}$.

Cite them by those names. Do not rename them. Do not put Hendrick's name on the other two. A later paper that restates one of these formulae is a restatement.

1. What is claimed, and what is not

Let $z_{m(t)} \in \mathbb{C}$ be the positions of N point vortices of circulations Γ_m , evolved by the 2π -periodic Biot–Savart law. A configuration is self-similar when there exist real A, B , independent of m , with

$$A + iB = \frac{i}{2\pi z_m} \sum_{n \neq m} \frac{\Gamma_n}{\bar{z}_m - \bar{z}_n}. \quad (1)$$

If $A < 0$, the vortices collide at the origin at the finite time $t_c = -\frac{1}{2A}$, rotating with instantaneous angular velocity ω so that $\omega_0 t_c = -\frac{B}{2A}$ is scale-invariant [1], [2], [3].

Claimed here. On three one-parameter exact families already in the literature, the product $\omega_0 t_c$ has a closed elementary form, and that form has a unique interior algebraic minimum. The minima, the equality configurations, and the names above.

Not claimed. The motion (Gröbli; Novikov–Sedov). The product as the pitch of a log spiral (Aref). The functions $A(\theta)$, $B(\theta)$ (Gotoda). The fastest-collapse time $t_c = 4\frac{\pi}{3}$ on Hendrick's

family at unit separation (Leoncini, Kuznetsov and Zaslavsky). Any published isolated lock listed in the search ledger of GENChase.

2. Hendrick's Identity

Family. $\Gamma = (1, 1, -\frac{1}{2})$, positions $z_1 = 0$, $z_2 = 1$, $z_3 = \frac{1}{2} + \left(\frac{\sqrt{3}}{2}\right)e^{i\theta}$ with $0 < \theta < \frac{\pi}{2}$. These lie on the angular-impulse circle $L = 0$ and collapse self-similarly [4]. The parameter θ is the third vortex's place on that circle, not an interior angle of the triangle.

$$\omega_0 t_c = \frac{2 - \cos^2 \theta}{\sin(2\theta)} \geq \sqrt{2}.$$

Equality uniquely on this arc at $\tan \theta = \frac{1}{\sqrt{2}}$, a triangle of interior angles 22.5° , 45° , 112.5° .

Proof of the minimum. Set $u = \tan \theta > 0$. Then $\omega_0 t_c = u + \frac{1}{2u}$ and

$$\omega_0 t_c - \sqrt{2} = \frac{(\sqrt{2}u - 1)^2}{2u} \geq 0.$$

At $\theta = 45^\circ$ the product is $\frac{3}{2}$. The same product splits, at this length, as

$$t_c = \left(\frac{\pi}{3}\right)\left(4u + \frac{1}{u}\right), \quad 2\pi\omega_0 = \frac{3(2u^2 + 1)}{4u^2 + 1},$$

which are Aref's τ and Ω in this family's angle [2]. Their product does not depend on the length unit. The collapse time itself has a unique minimum $4\frac{\pi}{3}$ at $u = \frac{1}{2}$; that number is already in Leoncini, Kuznetsov and Zaslavsky [5] and is not a second identity.

Name. Hendrick's Identity (also searched as Hendrick's law, Hendrick's Math). First public statement: GENChase, 19 September 2026, SharpMeow/GENChase#24.

3. Parallelogram lock

Family. $\Gamma = (1, 1, -2 - \sqrt{3}, -2 - \sqrt{3})$ at the vertices of a parallelogram whose diagonals meet at the origin, diagonal ratio $\frac{d_1}{d_2} = \sqrt{2 + \sqrt{3}}$, angle $\theta \in (0, \frac{\pi}{2})$ between the diagonals. These have $L = 0$ and collapse self-similarly [6]. The reciprocal pair $\Gamma = (1, 1, -2 + \sqrt{3}, -2 + \sqrt{3})$ with the diagonals swapped carries the same product.

$$\omega_0 t_c = \left(\frac{\sqrt{3}}{4}\right) \frac{4 - \cos 2\theta}{\sin(2\theta)} \geq \frac{3\sqrt{5}}{4}.$$

Equality uniquely on this arc at $\cos 2\theta = \frac{1}{4}$.

Proof of the minimum. Set $\varphi = 2\theta \in (0, \pi)$. The quotient $\frac{4 - \cos \varphi}{\sin \varphi}$ has a unique critical point on $(0, \pi)$ at $\cos \varphi = \frac{1}{4}$, where $\sin \varphi = \frac{\sqrt{15}}{4}$ and the quotient equals $\sqrt{15}$. Multiplying by $\frac{\sqrt{3}}{4}$ yields $3\frac{\sqrt{5}}{4}$. The sign of the first derivative on either side shows a minimum. Endpoints send the product to infinity. At $\theta = 45^\circ$ the product is $\sqrt{3}$. The product is $-\frac{B}{2A}$ from Gotoda's eq. (3.13) [3] specialized to this parallelogram; direct Biot-Savart on the 2π kernel matches that closed form to machine precision.

Name. Parallelogram lock. First public statement: GENChase, 20 September 2026, SharpMeow/GENChase#44. Not Novikov and Sedov’s t_* or ω separately under a new name.

4. Quincunx lock

Family. $\Gamma = (-1, -1, \frac{1}{2}, \frac{1}{2}, -\frac{3}{4})$, four vortices at the vertices of a parallelogram and one at the crossing of the diagonals, diagonal ratio $\frac{d_1}{d_2} = \frac{1}{\sqrt{2}}$, angle $\theta \in (0, \frac{\pi}{2})$ between the diagonals. These have $L = 0$ and collapse self-similarly [3], [6].

$$\omega_0 t_c = \left(\frac{3}{16} \right) \frac{7 - 4 \cos 2\theta}{\sin(2\theta)} \geq \frac{3\sqrt{33}}{16}.$$

Equality uniquely on this arc at $\cos 2\theta = \frac{4}{7}$.

Proof of the minimum. Set $\varphi = 2\theta \in (0, \pi)$. The quotient $\frac{7-4\cos\varphi}{\sin\varphi}\varphi$ has a unique critical point on $(0, \pi)$ at $\cos\varphi = \frac{4}{7}$, where $\sin\varphi = \frac{\sqrt{33}}{7}$ and the quotient equals $\sqrt{33}$. Multiplying by $\frac{3}{16}$ yields $3\frac{\sqrt{33}}{16}$. Endpoints send the product to infinity. At $\theta = 45^\circ$ the product is $\frac{21}{16}$. The product is $-\frac{B}{2A}$ from Gotoda’s five-vortex $A(\theta)$, $B(\theta)$ ($\gamma_3 \neq 0$); direct Biot–Savart matches. Gotoda plots this family as Hamiltonian against collapse rate and does not form the product or state its minimum [3], [7].

Name. Quincunx lock. First public statement: GENChase, 20 September 2026, SharpMeow/GENChase#44. Not the parallelogram lock. A different five-vortex Novikov–Sedov slice with diagonal ratio $\mu = 3$ recovers Hendrick’s product identically and is not a third identity.

5. Priority

The three statements were first published by Chaos in GENChase.

Statement	First public	Record
Hendrick’s Identity, $\omega_0 t_c \geq \sqrt{2}$	2026-09-19	SharpMeow/GENChase#24
Parallelogram lock, $\omega_0 t_c \geq 3\frac{\sqrt{5}}{4}$	2026-09-20	SharpMeow/GENChase#44
Quincunx lock, $\omega_0 t_c \geq 3\frac{\sqrt{33}}{16}$	2026-09-20	SharpMeow/GENChase#44

Canonical byte-exact lines and SHA-256 fingerprints live in `identities/STATEMENTS.txt` and `identities/HASHES.txt` of the same repository. Anyone may use the mathematics. Claiming first discovery of these closed forms or these floors, after those dates, without citing this note or `IDENTITIES.md`, is claiming this project’s work.

A later paper that restates one of these formulae is a restatement. Cite this note as the first public appearance unless an earlier dated source is produced. Tacchi’s Appendix B, which names related three-vortex coefficients, had not been opened from the machines that keep the search ledger at the time of this writing; if that appendix already states a row, that row will be withdrawn.

6. How to cite

Chaos, **Closed forms and sharp minima of ωt_c on three classical point-vortex collapse families**, GENChase (2026), <https://github.com/SharpMeow/GENChase>. Cite Hendrick's Identity, the parallelogram lock, and the quincunx lock by those names.

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  url         = {https://github.com/SharpMeow/GENChase},  
  note        = {Hendrick's Identity first public 2026-09-19;  
                parallelogram lock and quincunx lock 2026-09-20}  
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References

- [1] Y. Kimura, “Similarity solutions of two-dimensional point vortices,” *Journal of the Physical Society of Japan*, vol. 56, pp. 2024–2030, 1987.
- [2] H. Aref, “Self-similar motion of three point vortices,” *Physics of Fluids*, vol. 22, p. 57104, 2010, doi: [10.1063/1.3425649](https://doi.org/10.1063/1.3425649).
- [3] T. Gotoda, “Self-similar motions and related relative equilibria in the N -point vortex system”, *Journal of Dynamics and Differential Equations*, 2020.
- [4] W. Gröbli, “Spezielle Probleme über die Bewegung geradliniger paralleler Wirbelfäden,” 1877.
- [5] X. Leoncini, L. Kuznetsov, and G. M. Zaslavsky, “Motion of three vortices near collapse,” *Physics of Fluids*, vol. 12, p. 1911, 2000.
- [6] E. A. Novikov and Y. B. Sedov, “Vortex collapse,” *Soviet Physics JETP*, vol. 50, p. 297, 1979.
- [7] T. Gotoda, “Enstrophy variations in the collapsing process of point vortices,” *Journal of Fluid Mechanics*, 2025.